

VINAYKUMAR MUTHYALA

Big Data Engineer

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PROFESSIONAL SUMMARY

A highly skilled and innovative **Data Engineer** with over **5 years of experience** specializing in **Azure and AWS cloud ecosystems**. Demonstrates expertise in **data transformation, ETL pipeline development, and real-time data processing**. Proven ability to leverage **Azure Data Factory, Synapse, Databricks, and SQL** for building scalable solutions and enabling data-driven insights. Recognized for **streamlining workflows, reducing processing times by up to 30%**, and effectively utilizing **machine learning** to deliver actionable insights. Proficient in **Python, Spark, and Power BI**, and adept at collaborating across teams to solve complex data challenges.

EDUCATION

Masters in Computer Science, University of Central Missouri

Warrensburg, Missouri, USA (Aug 2021 - Dec 2022)

Bachelor of Technology in Mechanical Engineering

UBDT College of Engineering, Davanagere, Karnataka, India (Aug 2015 – Jun 2019)

SKILLS

- **Programming & Scripting:** Python, SQL, C++, Java, Scala
 - **Cloud Platforms:** Azure (Data Factory, Databricks, Synapse, Data Lake), AWS (S3, Redshift, Glue), GCP
 - **Big Data Technologies:** Hadoop, Spark, Kafka, Hive, Pig
 - **ETL/Analytics Tools:** Airflow, Power BI, Tableau, SSIS
 - **DevOps & CI/CD:** Azure DevOps, Jenkins, Docker, Kubernetes
 - **Machine Learning:** PyTorch, TensorFlow
 - **Operating Systems:** UNIX, Linux, Windows
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PROFESSIONAL EXPERIENCE

Data Engineer

Mercedes Benz, Georgia | June 2023 – Present

- Architected and managed **ETL pipelines** in Azure Data Factory (ADF), achieving a **30% improvement** in data processing speed.
- Designed robust **data models in Synapse Analytics** and created pipelines to optimize query performance and reduce latency by **20%**.
- Developed Azure DevOps CI/CD pipelines for **Databricks** deployments, enhancing efficiency and reducing errors in production environments.
- Developed Python-based automation scripts to optimize **ETL workflows** in **Azure Data Factory**, reducing manual intervention by **40%**.
- Built interactive **Power BI dashboards** to visualize insights for stakeholders, leading to faster, data-driven decisions.
- Worked on **Google cloud platform (GCP)** services like compute engine, cloud SQL.
- Developed custom **Python** scripts to automate **data ingestion and transformation**, reducing ETL processing time by **35%**.
- Developed, modified, and optimized complex **SQL queries and ETL processes**, reducing data processing time by **30%**.

Data Engineer

Cigna Healthcare, Connecticut | May 2022 – May 2023

- Automated **data ingestion processes** with Hadoop and Spark, significantly enhancing data processing speed for sensitive data analysis.
- Architected and implemented ETL frameworks using **Spark and Snowflake** to process large JSON datasets, improving data handling efficiency by **25%**.
- Built RESTful APIs with **AWS EC2**, achieving a scalable solution integrated with **S3** for data storage.
- Led a project to migrate SQL to **Azure Data Lake and Azure SQL Database**, eliminating 99% of data redundancy and achieving seamless data integration.
- Implemented data visualization and reporting using **Tableau** and automated data flows with Airflow for seamless **ETL operations**.
- Automated data ingestion and transformation workflows using **Python and Spark**, cutting down manual intervention by **40%**.
- Worked closely with Project Managers, Consultants, and internal teams to align technical solutions with **business requirements**.
- Conducted extensive **data pipeline optimizations** and parallel processing strategies, reducing ETL execution time by **30%**.

Associate Data Engineer

Sequential Technological International Pvt Ltd, Bengaluru , India | May 2019 – Apr 2021

- Developed and optimized Spark applications using **Scala and Spark SQL** to process high-volume data across varied formats, boosting efficiency by **40%**.
- Built **data ingestion workflows** in ADF for Azure Data Lake, improving ingestion rates and enabling faster insights for stakeholders.
- Created **Python-based** logging and error-handling mechanisms to enhance monitoring in **ETL** processes.
- Created robust data pipelines using **Azure Databricks** and SQL, supporting real-time and batch processing needs.
- Spearheaded migration efforts from legacy systems to cloud, including data transformation using **Hive and Spark SQL**.
- Collaborated with cross-functional teams to implement **Azure Synapse Analytics** for handling large-scale data, achieving **20% improvement in query performance**.
- Led a project to migrate SQL to **Azure Data Lake and Azure SQL Database**, eliminating 99% of data redundancy and achieving seamless data integration.

PROJECTS

E-commerce

- Developed a scalable system using **Python and SQL** to bookings payments, and customer service, improving data consistency and operational efficiency.
- Encoded and decoded json objects using **PySpark** to create and modify the data frames in Apache Spark

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (2023)
- Google Cloud Certified Professional Data Engineer (2023)